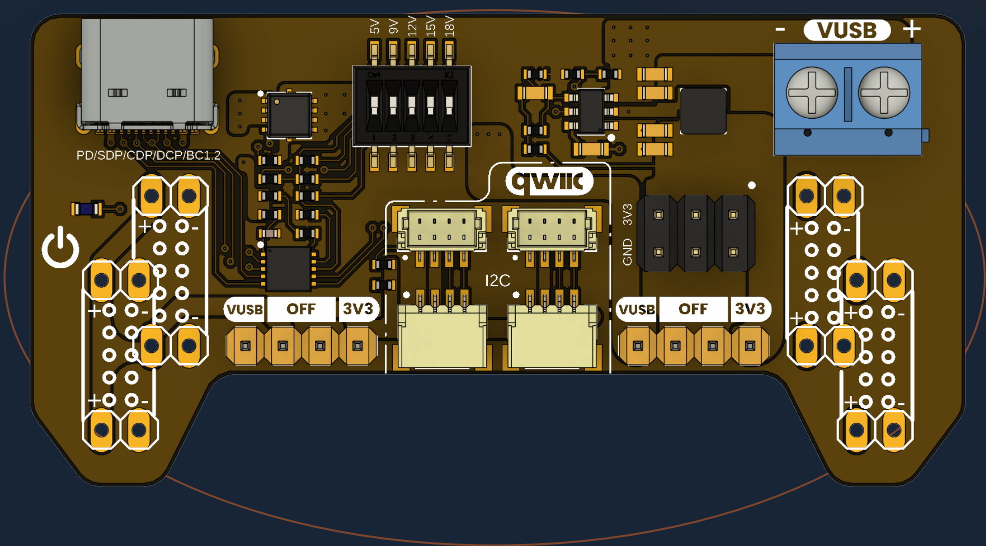
PRODUCT REFERENCE MANUAL

MEMBER OF THE UNIT DEVLAB ECOSYSTEM

Power Supply for Breadboard with Qwiic

Version 1.0.0



USB-C Power Delivery sink and breadboard power-distribution module with selectable VUSB rails, regulated 3.3 V output, and Qwiic I²C expansion.

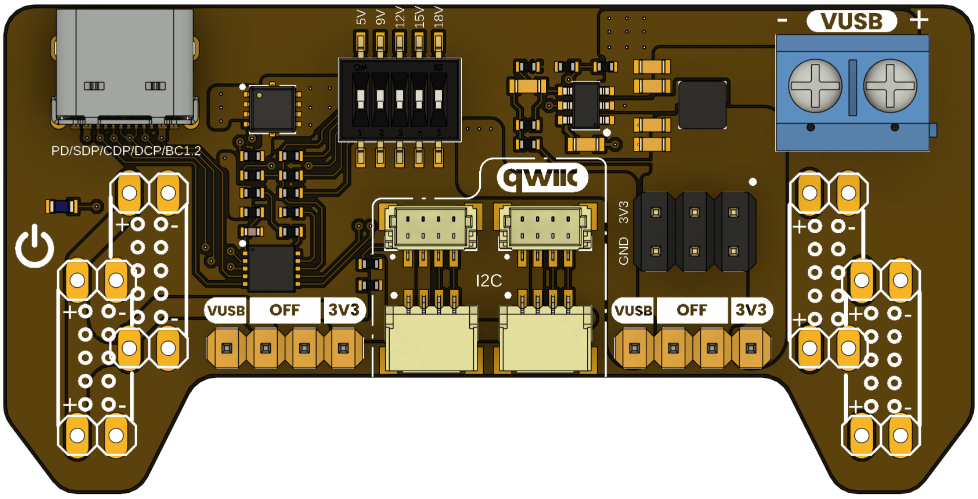
USB-C PD SINK | 3.3 V / 2 A | UP TO 20 V PD | QWIIC · I²C

DOCUMENT MAP

Contents

Description

The DevLab Power Supply for Breadboard with Qwiic is a USB Type-C-powered power-distribution module for breadboard-based electronic prototyping. It acts as a USB Power Delivery (PD) sink, negotiates a profile from a compatible source, and routes the negotiated VUSB rail to selectable breadboard outputs. An onboard TPS54302 buck regulator supplies a separate 3.3 V rail.



Power Supply for Breadboard with Qwiic top view

Applications

- Breadboard power distribution and mixed-voltage prototyping
- USB-C PD source and cable evaluation
- Embedded-system, sensor, and peripheral development
- Laboratory education, validation, and early hardware testing

Hardware Features

- USB Type-C PD input with HUSB238 PD-sink controller
- Selectable VUSB profiles supported by the attached PD source
- TPS54302-based 3.3 V output rail
- Independent left and right breadboard-rail selection
- Screw-terminal and header outputs for VUSB, 3.3 V, and ground
- Qwiic-compatible I^2C connectors for 3.3 V peripherals
- DIP-switch PD configuration and optional HUSB238 I^2C access

The module is mechanically intended for common 54 mm and 64 mm breadboard rail formats. Output voltage and available current depend on the negotiated PD profile, source, cable, thermal conditions, and external load.

1 The Board

The Power Supply for Breadboard with Qwiic provides a compact power entry and distribution point for solderless breadboards. A USB-C PD source supplies the module. The selected VUSB rail and the regulated 3.3 V rail can be assigned to the two breadboard sides independently, which supports both single-voltage and mixed-voltage prototypes.

1.1 Board Identification

Item	Value
Product	Power Supply for Breadboard with Qwiic
Manufacturer part number	UE0113
Product family	UNIT Electronics DevLab
Product type	USB-C PD breadboard power-distribution module
Hardware revision	V1.0
Product Reference	Version 1.0.0
PD controller	HUSB238
3.3 V regulator	TPS54302

1.2 Included and Recommended Items

The released V1.0 datasheet lists the following accessory items: 16 single-row male headers, one Qwiic 4-pin JST 1.0 mm harness, and two jumper caps.

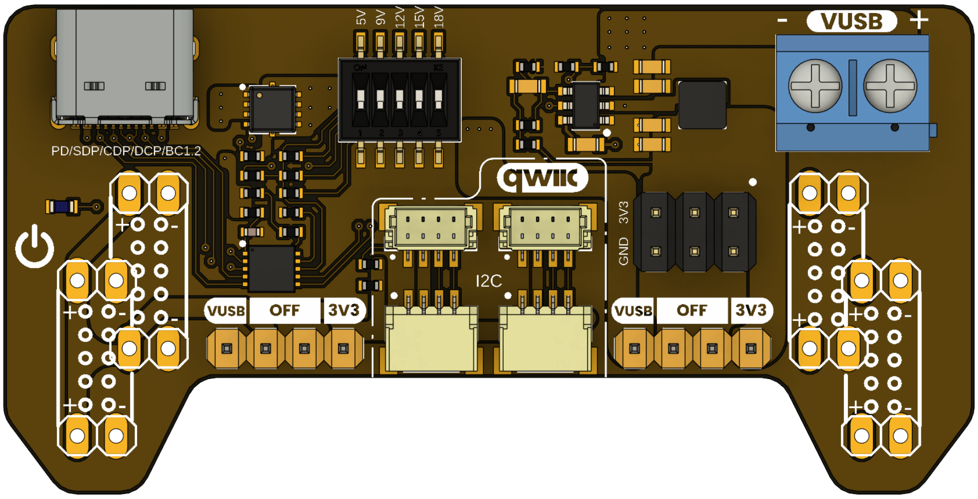
For safe operation, use a USB-C charger that supports the needed PD profile, a USB-C cable rated for the expected voltage and current, and a breadboard with the appropriate rail spacing. Verify the source profile before connecting a load.

1.3 Main Assemblies

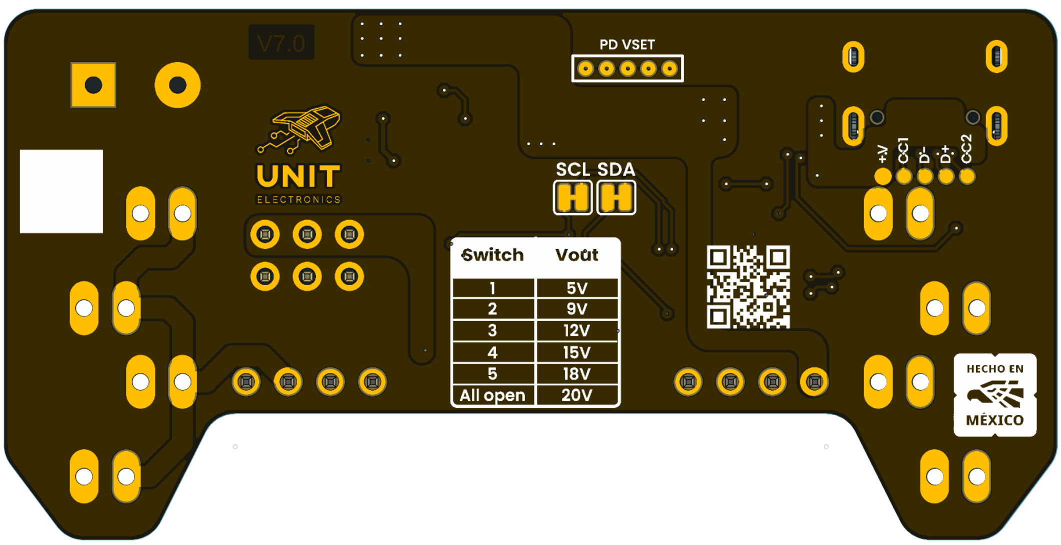
RefDes	Component or connector	Function
J1	USB Type-C connector	PD power input
IC1	HUSB238	PD negotiation controller
SW1	DIP switch	PD voltage-profile selection
U1	TPS54302	3.3 V buck regulator
L1	Power LED	Power indication
J6	Screw terminal	VUSB output access
JP7	Pin header	3.3 V output access
JP3 / JP4	Selector headers	Left and right rail-voltage selection
JP1 / JP2	Left headers	Breadboard power connection

RefDes	Component or connector	Function
JP5 / JP6	Right headers	Breadboard power connection

1.4 Board Views



Top view



Bottom view

1.5 Handling

Disconnect USB-C power before moving rail-selection jumpers, inserting the board into a breadboard, or changing wiring. Keep high-voltage VUSB wiring away from the 3.3 V logic rail and confirm polarity before connecting a load.

2 Ratings

The following values are migrated from the released V1.0 product datasheet. They describe intended module operation and do not replace the limits in the HUSB238 or TPS54302 component datasheets.

2.1 Recommended Operating Conditions

Parameter	Description	Min	Typ	Max	Unit
VIN	USB Type-C PD input voltage	5	—	20	V
VBUS	USB PD negotiated output voltage	5	9 / 12 / 15	20	V
V3V3	Regulated 3.3 V output rail	3.2	3.3	3.4	V
I3V3	3.3 V regulator output current	—	—	2	A
IBUS	Breadboard output current	—	—	5	A
VIH	I²C high-level input voltage	0.7 × VCC	—	5.5	V
VIL	I²C low-level input voltage	—	—	0.3 × VCC	V
ICC	Module supply current	—	3.1	—	mA
fI2C	I²C communication frequency	0	100	400	kHz
TOPD	USB PD supported profiles	5	9 / 12 / 15	20	V
TAMB	Recommended ambient temperature	-20	25	85	°C

2.2 Electrical Notes

1. PD voltage can be selected through the onboard DIP-switch resistor network or dynamically through the HUSB238 I²C interface. I²C configuration takes precedence over DIP-switch selection.
2. The 3.3 V output capability depends on input voltage, thermal dissipation, board airflow, and load conditions.
3. Qwiic/STEMMA QT connectors share 3.3 V, ground, SDA, and SCL for daisy-chain operation.
4. VBUS output current depends on the available source profile, cable, and electrical characteristics of the attached PD power adapter.
5. For stable 3.3 V regulation, use a 9 V, 12 V, 15 V, or 20 V PD profile. At 5 V, the input must remain above 4.5 V under load to avoid TPS54302 UVLO.

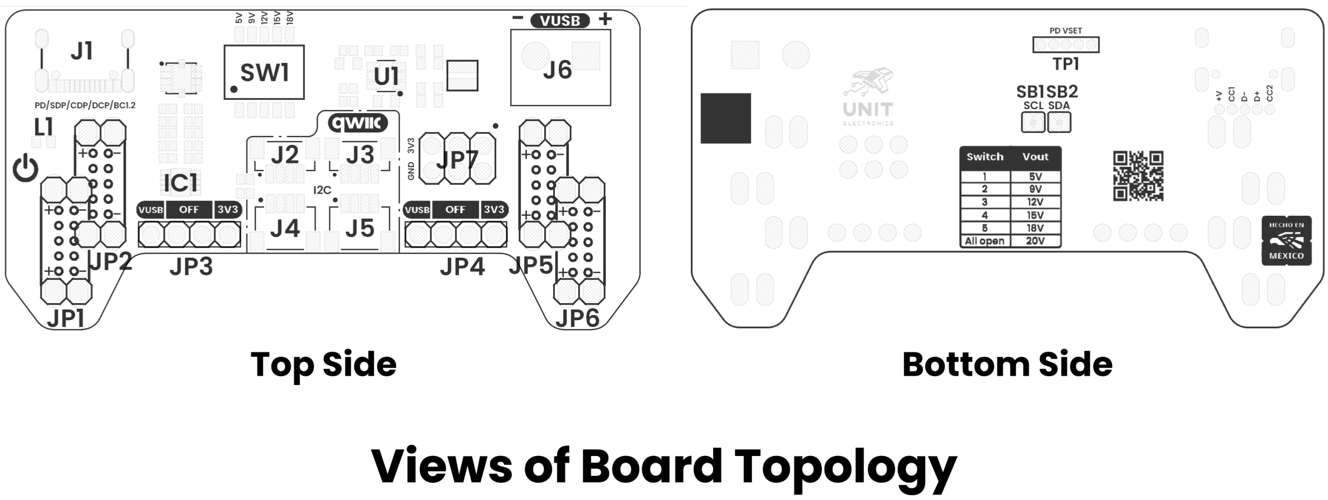
2.3 Power-Domain Scope

VUSB is the negotiated USB-C PD output rail. 3V3 is generated by the TPS54302 and is intended for low-voltage logic and Qwiic peripherals. Both rails share ground but must not be treated as interchangeable. Available current for a specific setup is limited by the PD source, cable, selected profile, onboard consumption, temperature, and connected load.

3 Functional Overview

The module receives power from a USB-C PD source, negotiates a compatible fixed profile through the HUSB238 controller, and distributes that negotiated VUSB rail to the screw terminal and selectable breadboard headers. In parallel, the TPS54302 converts VUSB to a regulated 3.3 V rail for logic and Qwiic peripherals.

3.1 Power Distribution Architecture



Board topology

The left and right breadboard rails have independent selectors. Each selector can route VUSB, disconnect the rail, or route 3.3 V. This makes it possible to use a higher PD voltage for one rail while retaining 3.3 V logic on the other.

Selection	Function
VUSB	Routes the negotiated USB-C PD voltage to the rail
OFF	Disconnects the rail output
3V3	Routes the regulated 3.3 V rail to the rail

3.2 USB-C Power Delivery

The HUSB238 performs USB-C PD-sink negotiation with a compatible source. The available profiles are determined by the charger and cable; selecting a profile does not create a voltage that the source did not advertise. The released datasheet describes 5 V, 9 V, 12 V, 15 V, and 20 V as the standard profiles, and also mentions 18 V where that profile is offered by the source.

Use the onboard DIP switch for standalone configuration. When HUSB238 I²C access is enabled, a host controller can monitor the connection and apply dynamic selection; I²C configuration has priority over the DIP-switch setting.

3.3 Qwiic Expansion

The Qwiic-compatible JST 1.0 mm connectors expose a shared 3.3 V I²C bus: SDA, SCL, 3.3 V, and ground. Optional solder jumpers connect the HUSB238 to that bus for monitoring and control. Leave those jumpers open when the controller must remain isolated from the external Qwiic network.

The I²C bus is a 3.3 V logic domain. Do not apply VUSB to SDA, SCL, or the Qwiic power pin.

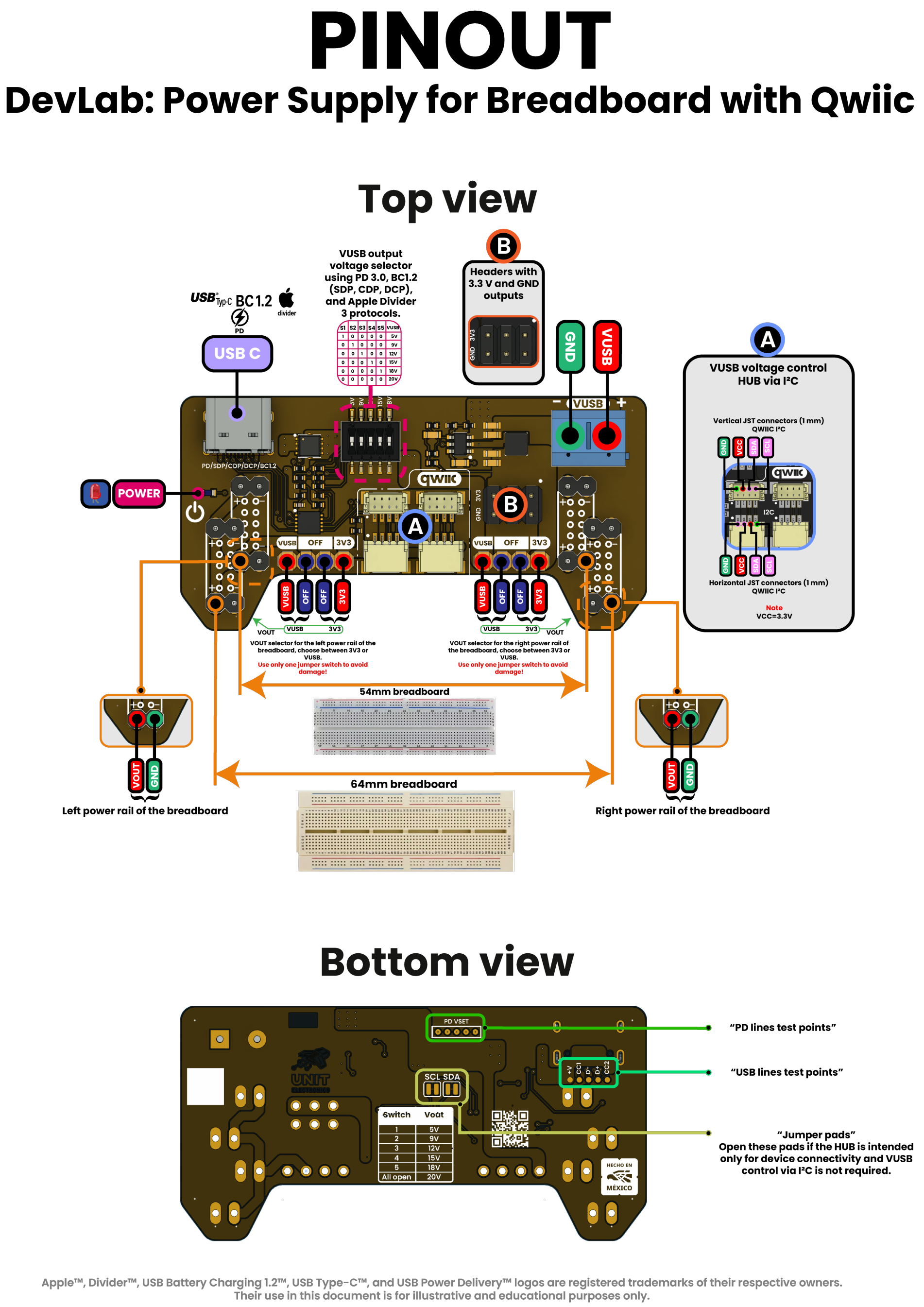
3.4 Breadboard Integration

The mechanical layout aligns with common 54 mm and 64 mm breadboard rail formats. The two sides permit independent power assignment for cases such as:

Left rail	Right rail	Typical use
3.3 V	3.3 V	Logic and sensor prototype
VUSB	3.3 V	Mixed power and logic system
12 V	3.3 V	Motor driver with microcontroller
OFF	VUSB	Single high-voltage rail test

The 12 V example assumes that the source provides and the module has selected that VUSB profile.

4 Connectors and Pinout



General pinout

4.1 General Pinout Description

Label	Function	Notes
VUSB	USB PD output rail	Negotiated PD voltage; 5 V to 20 V according to the attached source and selected profile
3V3	Regulated output	Generated by the TPS54302 regulator

Label	Function	Notes
GND	Ground reference	Common reference for power and signals
SDA	I^2C data	Qwiic/STEMMA QT bus; optional HUSB238 access
SCL	I^2C clock	Qwiic/STEMMA QT bus; optional HUSB238 access
USB-C	PD input	Power input from a compatible USB-C PD source
DIP SW	PD configuration	Selects the standalone PD request configuration
Qwiic	I^2C connector	JST 1.0 mm, 3.3 V peripheral interface
VOUT	Breadboard power output	Selected power rail for external circuits
GND OUT	Breadboard ground	Ground output for external circuits
Header A	Breadboard interface	Power-rail connection
Header B	Auxiliary power header	VUSB, 3.3 V, and ground outputs

4.2 Qwiic I^2C Connector

Pin	Voltage level	Function
VCC	3.3 V module rail	Power for a connected 3.3 V-compatible peripheral
GND	0 V	Common ground
SDA	1.8 V to VCC	I^2C serial data
SCL	1.8 V to VCC	I^2C serial clock

The board’s Qwiic/STEMMA QT connectors share one 3.3 V SDA/SCL bus and are intended for daisy-chain connectivity. Use one voltage-selection jumper configuration per rail; conflicting selections can damage the module or connected hardware.

4.3 Output Connections

The screw terminal exposes VUSB for a direct external connection. The breadboard headers distribute the voltage selected by their corresponding left or right selector. Disconnect power before changing selector jumpers or moving the module between breadboards.

5 Board Operation

5.1 Initial Power-Up

1. With USB-C disconnected, set both breadboard-rail selectors to `OFF`.
2. Insert the module in the intended breadboard position and inspect that each header aligns with the intended power rail.
3. Connect a compatible USB-C PD source and cable.
4. Confirm the negotiated VUSB profile before connecting a load.
5. Set each rail independently to `VUSB`, `3V3`, or leave it `OFF`.
6. Verify voltage and polarity with a meter before connecting sensitive or high-current circuitry.

5.2 Standalone PD Selection

The onboard DIP switch configures the HUSB238 request when no external I²C controller overrides it. Use only profiles supported by the connected source. An unavailable request must not be assumed to result in the selected voltage; measure VUSB before use.

5.3 I²C Monitoring with Arduino

When the optional HUSB238-to-Qwiic connections are enabled, an external 3.3 V microcontroller can access the controller through I²C. The released example uses an ESP32-compatible controller with SDA on GPIO6, SCL on GPIO7, and the default HUSB238 address `0x08`.

Install the `Adafruit_HUSB238` library and use the supplied example:

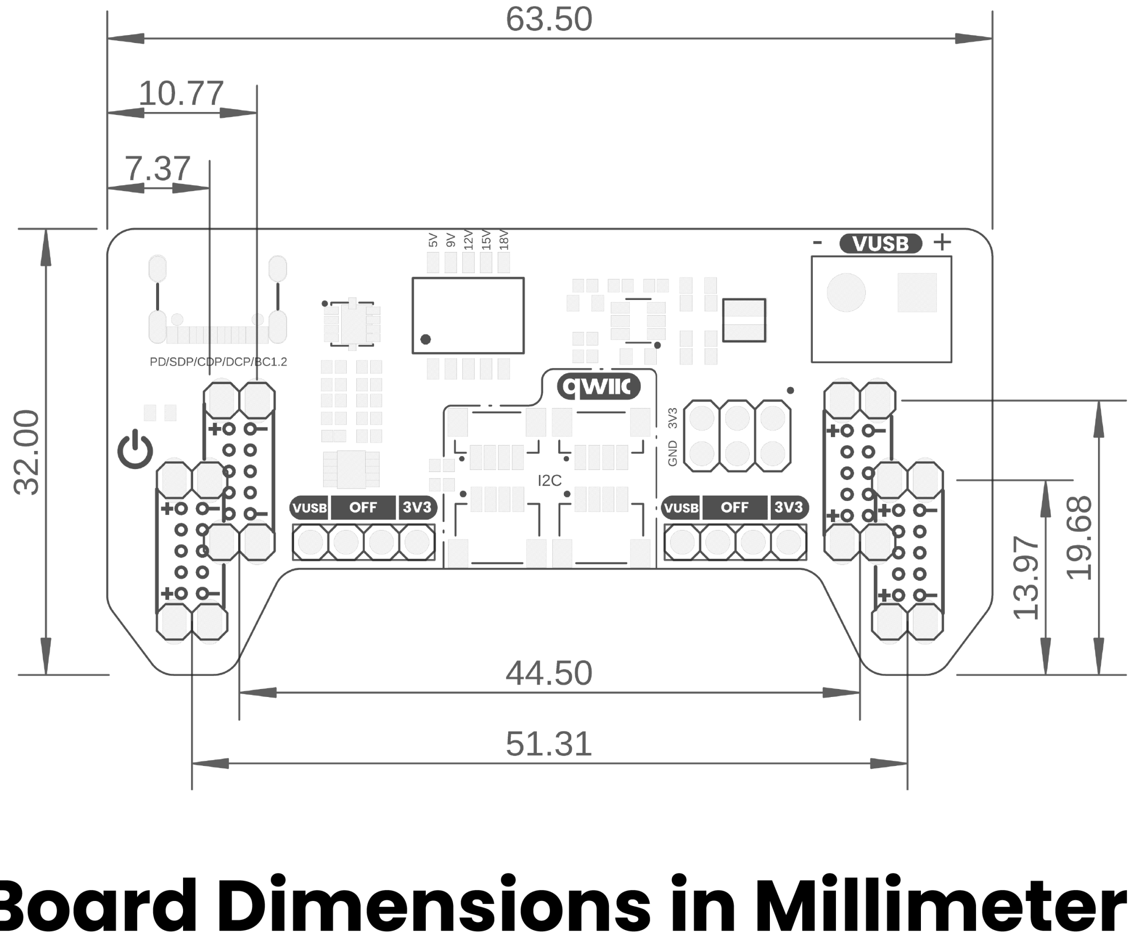
```
software/examples/cpp_examples/husb238_detection_only/husb238_detection_only.ino
```

The example initializes I²C, detects the HUSB238, and reports the PD profiles advertised by the attached source. It is a detection example; it does not validate the connected load or replace a voltage measurement.

5.4 Operational Precautions

- Do not change rail jumpers while the module is powered.
- Keep the VUSB rail isolated from 3.3 V-only devices and I²C signals.
- Use a cable and PD source rated for the requested voltage and expected load.
- The maximum usable output current is installation-dependent; respect source, cable, regulator, thermal, breadboard, and connected-device limits.
- For a 3.3 V load, prefer a 9 V or higher PD input profile as documented in the electrical notes.

6 Mechanical Information



Mechanical dimensions in millimeters

All dimensions in the drawing are in millimeters. The board is designed to align with common 54 mm and 64 mm solderless-breadboard rail formats. Confirm actual clearance around the USB-C connector, screw terminal, jumper caps, and Qwiic cable before final assembly.

6.1 Installation Notes

Support the board while inserting or removing it from a breadboard to avoid bending headers. Do not use the Qwiic cable or USB-C cable as a mechanical support. Keep adequate clearance for wiring attached to the VUSB terminal and for heat dissipation from the regulator under load.

7 Company Information

Item	Value
Company	UNIT Electronics
Website	https://uelectronics.com/
Address	Salvador 19, Cuauhtémoc, 06000, Mexico City, CDMX, Mexico
Product family	DevLab

For product support, current documentation, and software examples, use the product wiki and repository listed in the next chapter.

8 Reference Documentation

Reference	Link or repository file
Product wiki	Power Supply for Breadboard with Qwiic wiki
Source repository	UNIT-Electronics-MX/ unit_devlab_power_supply_for_breadboard_with_qwiic
Board schematic	V1.0 schematic
General pinout	English pinout
HUSB238 controller	`hardware/resources/ unit_datasheet_v_1_0_0_ue0113_husb238.pdf`
TPS54302 regulator	`hardware/resources/ unit_datasheet_v_1_0_0_ue0113_tps54302.pdf`
Arduino IDE	arduino.cc/en/software

Component datasheets describe the individual devices. The ratings and mechanical information in this reference remain specific to the UE0113 module.

9 Appendix

9.1 Source Migration

This editable product reference is migrated from the released PDF:

`hardware/unit_datasheet_v_1_0_0_ue0113_Power_Supply_for_Breadboard_with_Qwiic.pdf`

The original PDF is retained as a released historical artifact. The Markdown chapters under `tools/product-reference/chapters/` are the source for future generated PDF and DOCX editions.

9.2 Documentation Scope

The module's negotiated VUSB output depends on the USB-C PD source and cable. The listed component ratings do not independently establish all module-level limits. Before high-voltage or high-current use, verify the selected profile, measure the output, and account for all connected hardware.

The released V1.0 datasheet mentions an 18 V PD profile in descriptive text, while its ratings tables enumerate 5 V, 9 V, 12 V, 15 V, and 20 V. This reference treats 18 V as source-dependent rather than as a guaranteed standard profile.

9.3 Document Control

Item	Value
Product reference version	1.0.0
Hardware revision	V1.0
Manufacturer part number	UE0113
Released datasheet date	2026-06-30
Editable source	`tools/product-reference/`